Linear Algebra II HW 9

Due Date: on or before 14:00, April/30/2025 (Wednesday), in T3-502-R26-H7.

1. Find an orthogonal or unitary diagonalizing matrix for each of the following:

(a) $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 3+i \\ 3-i & 4 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & i & 0 \\ -i & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

2. Consider $A = \begin{pmatrix} 16a-12 & 12a-9 \\ 12a+16 & 9a+12 \end{pmatrix}$.

(a) Show that $\det(A) = 0$ and find an eigenvector of A corresponding to eigenvalue 0.

(b) Suppose the algebraic multiplicity of 0 $>$ geometric multiplicity of 0. Show that $a = 0$. Find an orthogonal matrix Q and an upper triangular matrix T such that $A = QTQ^T$.

3 Consider $-3x^2 + 6xy + 5y^2 - 24 = 0.$

(i) Find a suitable change of coordinates (i.e., a rotation and/or a translation) so that the resulting conic section is in standard form,

(ii) Identify the curve.

4. Let A be a 2×2 matrix such that $A = UTU^H$ with unitary matrix U and upper triangular matrix T. Suppose $t_{12} \neq 0$. Show that

(a) the eigenvalues of A are $\lambda_1 = t_{11}$ and $\lambda_2 = t_{22}$.

(b) $\boldsymbol{u}_1$, the first column of U, is an eigenvector of A belonging to $\lambda_1 = t_{11}$.

(c) $\boldsymbol{u}_2$, the second column of U, is not an eigenvector of A belonging to $\lambda_2 = t_{22}$.

5. Show that if a matrix U is both unitary and Hermitian, then any eigenvalue of U must equal either 1 or -1.